

PENETRATION TESTING & SECURITY ASSESSMENT REPORT

Infrastructure Vulnerability Audit & Exploitation Proof of Concept

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Target IP: **192.168.56.103**
Assessment Status: **COMPROMISED (CRITICAL)**

EXECUTIVE SUMMARY

4 / 4

MISSIONS COMPLETED

CRITICAL

OVERALL RISK LEVEL

2

FLAGS EXFILTRATED

FULL

SYSTEM ACCESS

A comprehensive technical penetration testing engagement was executed against target system **192.168.56.103**. The objective was to identify security flaws, evaluate exploitation vectors, and retrieve proof-of-concept evidence across network services, web applications, and SMB shares. All four predefined missions were successfully executed, revealing critical misconfigurations and code injection vulnerabilities.

MISSION	TARGET VECTOR / SERVICE	VULNERABILITY CLASS	RISK LEVEL	STATUS
Mission 01	Port 21 (vsFTPd 3.0.3)	Unauthenticated Anonymous Access	MEDIUM	COMPLETED
Mission 02	Port 80 (/ping.php)	OS Command Injection (RCE)	CRITICAL	COMPLETED
Mission 03	Port 80 (/admin/)	Control Panel Privilege Bypass	HIGH	COMPLETED
Mission 04	Port 445 (SMB Share)	Anonymous SMB Share Exposure	HIGH	COMPLETED

TARGET ATTACK CHAIN DIAGRAM

Target Architecture: 192.168.56.103 (EV Cyber Academy Host)

Mission 1

FTP Anonymous
flag.txt Exfiltrated

Mission 2

RCE (ping.php)
www-data Context

Mission 3

Admin Portal
Central Control Granted

Mission 4

SMB Null Session
confidential_flag.txt

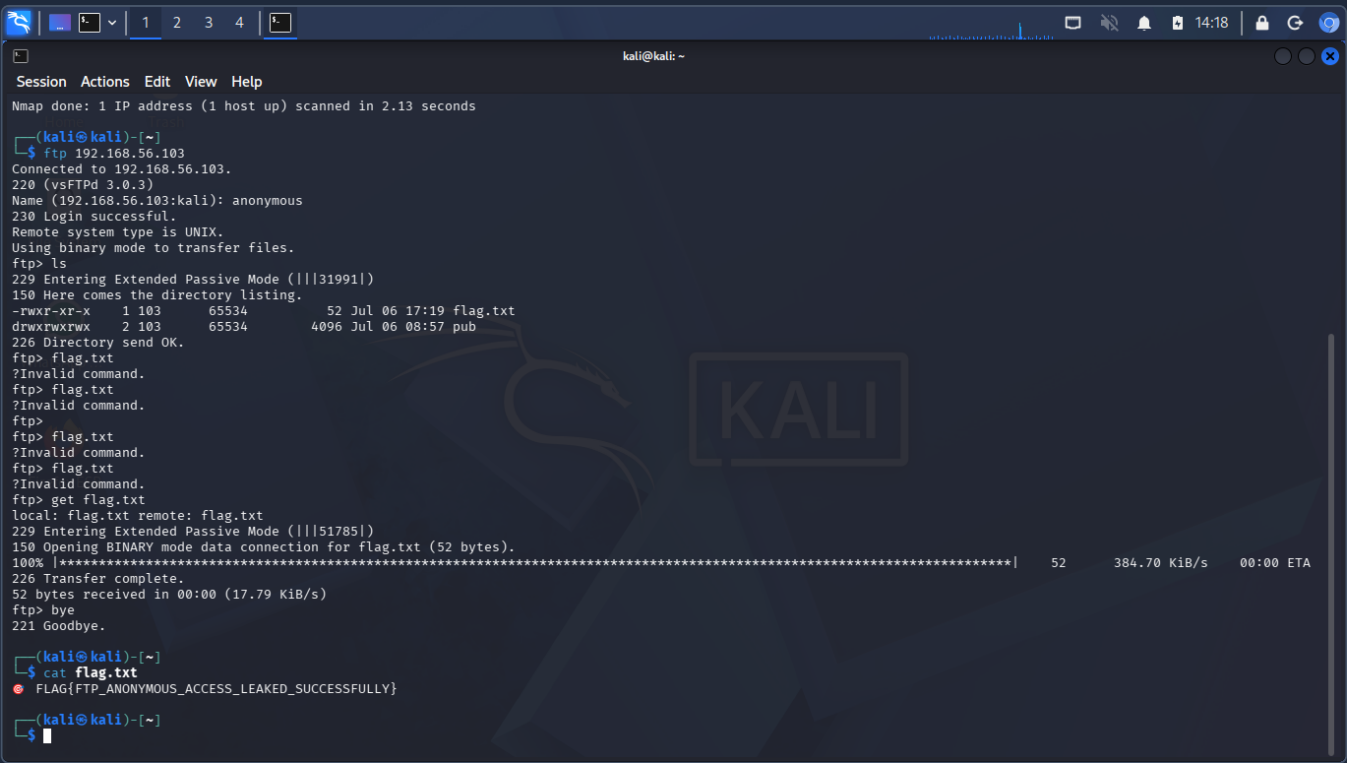
DETAILED EVIDENCE & EXPLOITATION MISSIONS

Mission 1: Anonymous FTP Data Exfiltration

MEDIUM RISK

Analysis & Findings: An FTP service running vsFTPD 3.0.3 on port 21 was identified allowing unauthenticated anonymous logins. Directory listing yielded flag.txt, which was successfully downloaded locally via binary transfer.

```
$ ftp 192.168.56.103
220 (vsFTPD 3.0.3)
Name: anonymous | 230 Login successful.
ftp> get flag.txt
$ cat flag.txt
```



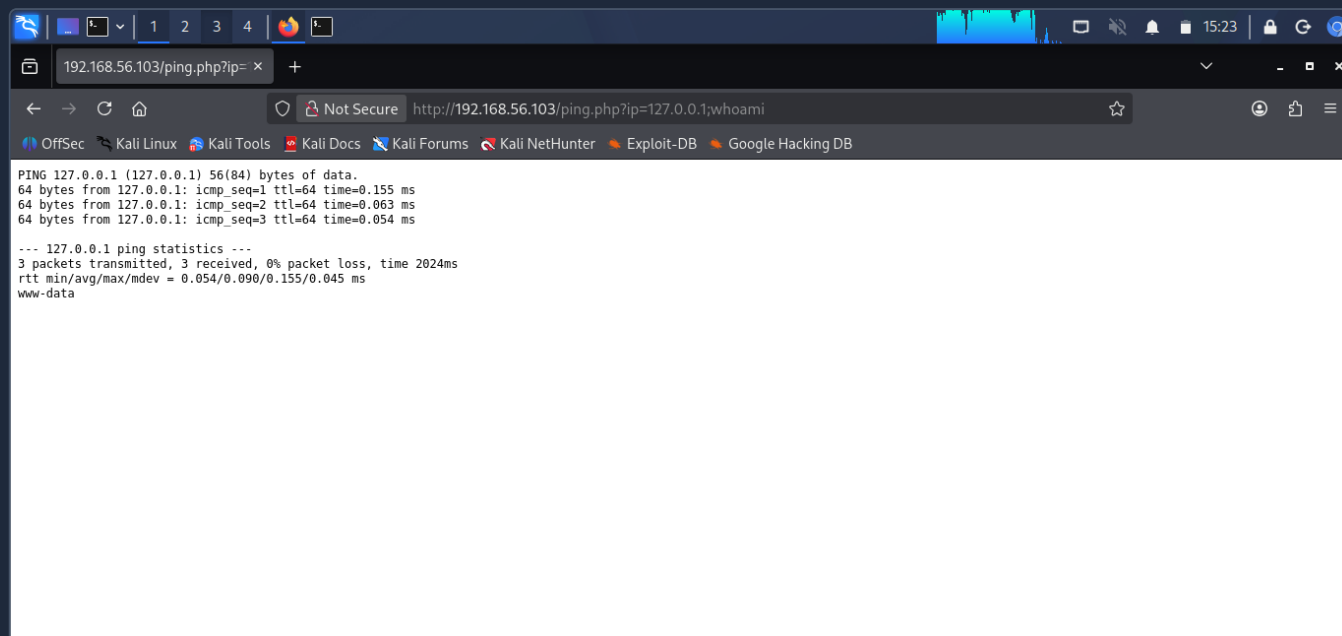
Exfiltrated Flag: FLAG{FTP_ANONYMOUS_ACCESS_LEAKED_SUCCESSFULLY}

Mission 2: Web Command Injection (Remote Code Execution)

CRITICAL RISK

Analysis & Findings: The web endpoint `/ping.php` fails to sanitize input passed to the `ip` parameter. By inserting semicolon operators (`;`), arbitrary system commands were executed on the underlying Linux host under the web server user account (`www-data`, `uid=33`).

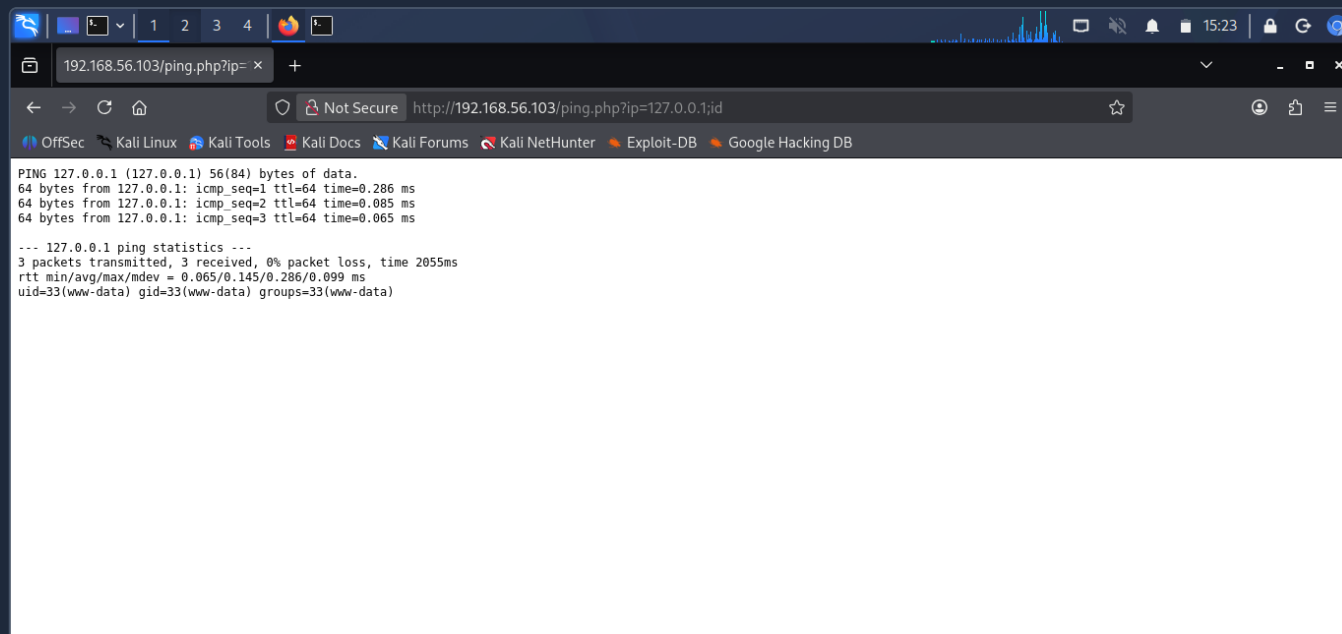
Execution 1: Context Verification (whoami)



```
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.155 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.063 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.054 ms

--- 127.0.0.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2024ms
rtt min/avg/max/mdev = 0.054/0.090/0.155/0.045 ms
www-data
```

Execution 2: UID / GID Enumeration (id)



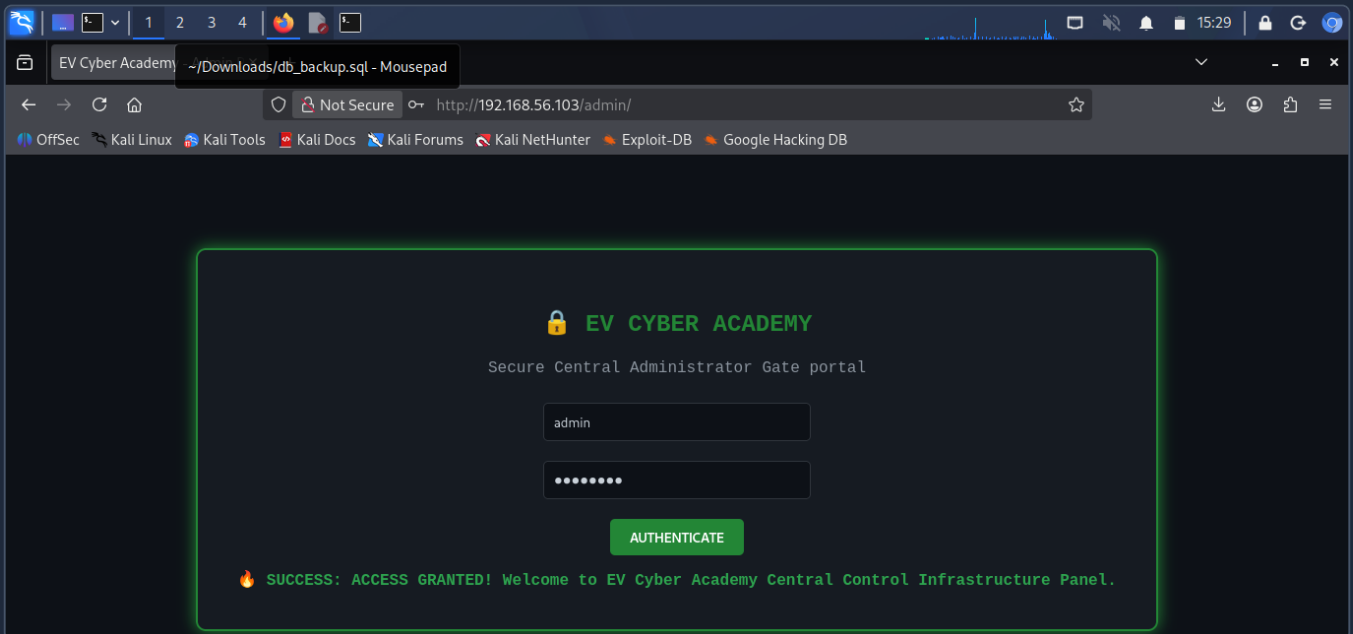
```
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.286 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.085 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.065 ms

--- 127.0.0.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2055ms
rtt min/avg/max/mdev = 0.065/0.145/0.286/0.099 ms
uid=33(www-data) gid=33(www-data) groups=33(www-data)
```

Mission 3: Central Administrative Gate Portal Access

HIGH RISK

Analysis & Findings: Access was gained to the EV Cyber Academy Central Control Infrastructure Panel located at /admin/. Authenticating with administrative credentials granted immediate access to the central control infrastructure gateway.



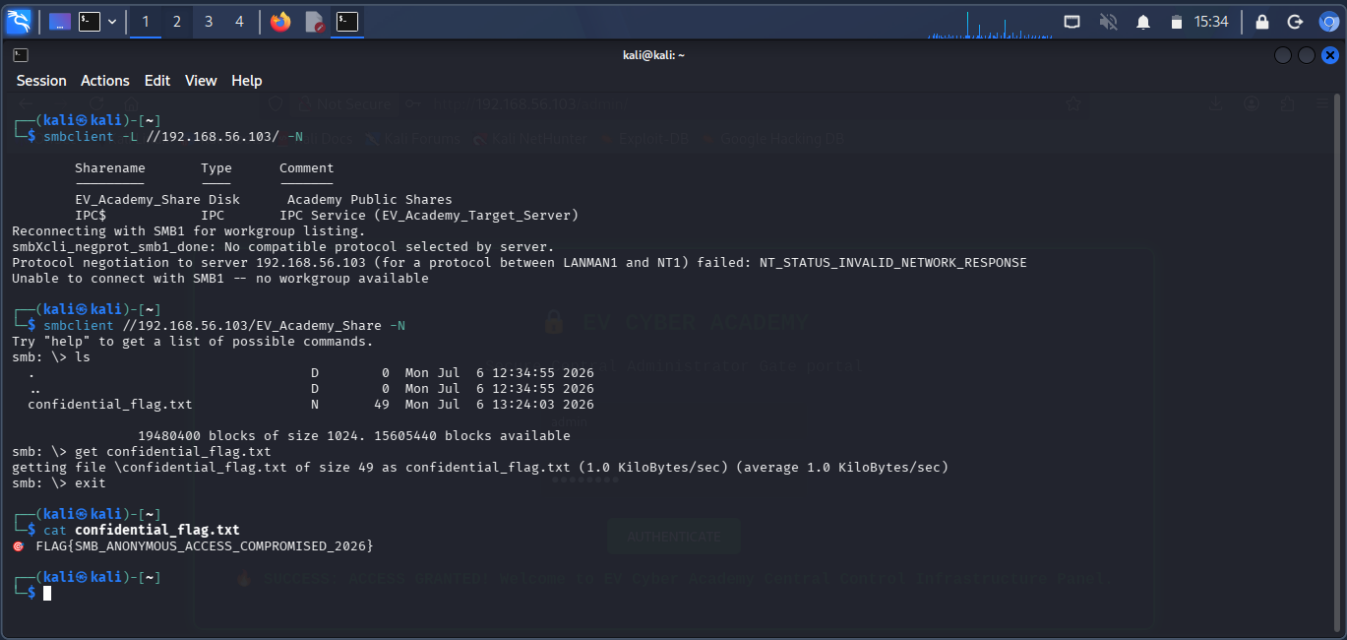
SUCCESS: ACCESS GRANTED! Welcome to EV Cyber Academy Central Control Infrastructure Panel.

Mission 4: Unauthenticated SMB Share File Exfiltration

HIGH RISK

Analysis & Findings: Enumeration of port 445 via SMB client revealed an unauthenticated share named EV_Academy_Share. Accessing the share via a null session (-N) allowed reading and exfiltrating confidential_flag.txt.

```
$ smbclient //192.168.56.103/EV_Academy_Share -N
smb: \> get confidential_flag.txt
$ cat confidential_flag.txt
```



Exfiltrated Flag: FLAG{SMB_ANONYMOUS_ACCESS_COMPROMISED_2026}

REMEDIATION RECOMMENDATIONS

PRIORITY	VULNERABLE VECTOR	RECOMMENDED ACTION
P0	ping.php Command Injection	Sanitize and validate input using strict regex or use native library functions instead of system shell execution commands.
P1	SMB Null Session Share Access	Disable anonymous/guest access in Samba configuration file (smb.conf) and enforce authentication.
P1	FTP Anonymous Read Access	Disable anonymous FTP access by setting anonymous_enable=NO in /etc/vsftpd.conf.
P2	Administrative Portal Access	Implement Multi-Factor Authentication (MFA) and restrict access to ^/admin^ via IP whitelist / VPN.